

mux

July 31, 2025

```
[47]: from main import peaks_vs_thresholds
import numpy as np
import matplotlib.pyplot as plt
from natsort import natsorted
from glob import glob
import re
import figures.functions as ft

%matplotlib widget
```

```
[48]: pulse_models = {}
```

```
[49]: dir = r"D:\Data\LT218Chip1_BF_20221103_MIR3_8\12KIDs mono on 3800 long\TD_Power"
info_files = natsorted(glob(dir + '\*TDvis0*_info.dat'))
kid_table = np.zeros((27, 2))
kid_table[:, 0] = np.arange(27)
f0s = np.load(r'D:\Data\LT218_MUX\LT218_27f0s_from_VNA.npy')
for file in info_files:
    id = int(re.findall('KID\d+', file)[0][3:])
    f0_meas = ft.get_info(file)[0]
    id_main = np.argmin(np.absolute(f0_meas - f0s))
    if kid_table[id_main,1] != 0:
        print(f"Warning: KID{kid_table[id_main,1]} with f0 {f0s[id_main]}_
↳overwritten by KID{id} with f0 {f0_meas}")
        kid_table[id_main,1] = id
    # print(id_main, id)
kid_table[25,1] = 23
kid_table
```

Warning: KID23.0 with f0 4.6039375 overwritten by KID24 with f0 4.604713

```
[49]: array([[ 0.,  0.],
            [ 1.,  0.],
            [ 2.,  0.],
            [ 3.,  0.],
            [ 4.,  0.],
            [ 5.,  3.],
            [ 6.,  0.]
```

```

[ 7.,  5.],
[ 8.,  0.],
[ 9.,  0.],
[10.,  8.],
[11.,  9.],
[12., 10.],
[13.,  0.],
[14., 13.],
[15.,  0.],
[16.,  0.],
[17., 16.],
[18., 17.],
[19.,  0.],
[20.,  0.],
[21.,  0.],
[22., 20.],
[23., 21.],
[24.,  0.],
[25., 23.],
[26., 24.]]))

```

```

[53]: kid = 20
      pread = 108
      file_type = 'vis'
      pw = 300
      pw_offset = 20
      filter_type = 'exp'
      lifetime = 40
      fit_tqp = [50, 300]

      dir_on = r"D:\Data\LT218Chip1_BF_20221103_MIR3_8\12KIDs mono on 3800_
      ↪long\TD_Power"
      chunksize = 25
      noise_mph = 5
      pulse_mph = 16
      doubles_fraction = 10
      nr_stds = [16]
      nr_chuncks = [None]
      pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,
      ↪pulse_mph, pw, pw_offset, lifetime, nr_stds,
      ↪nr_chuncks, chunksize, filter_type, fit_tqp,
      ↪doubles_fraction=doubles_fraction)

      kid_id = np.argwhere(kid_table[:, 1] == kid)[0][0]
      name = 'KID%d' % kid_id
      pulse_models[name] = {}
      pulse_models[name]['pulse template'] = np.mean(pulses, axis=0)

```

Secondaries/Outliers/Total = 20/7/276

Optimal filter constructed with 251 pulses and 1000 noise segments

c:\Users\wilbertr\ownCloud2\PhD\PhD\PythonProjects\mir-kids\main.py:363:

RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot interface (``matplotlib.pyplot.figure``) are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam ``figure.max_open_warning``). Consider using ``matplotlib.pyplot.close``.

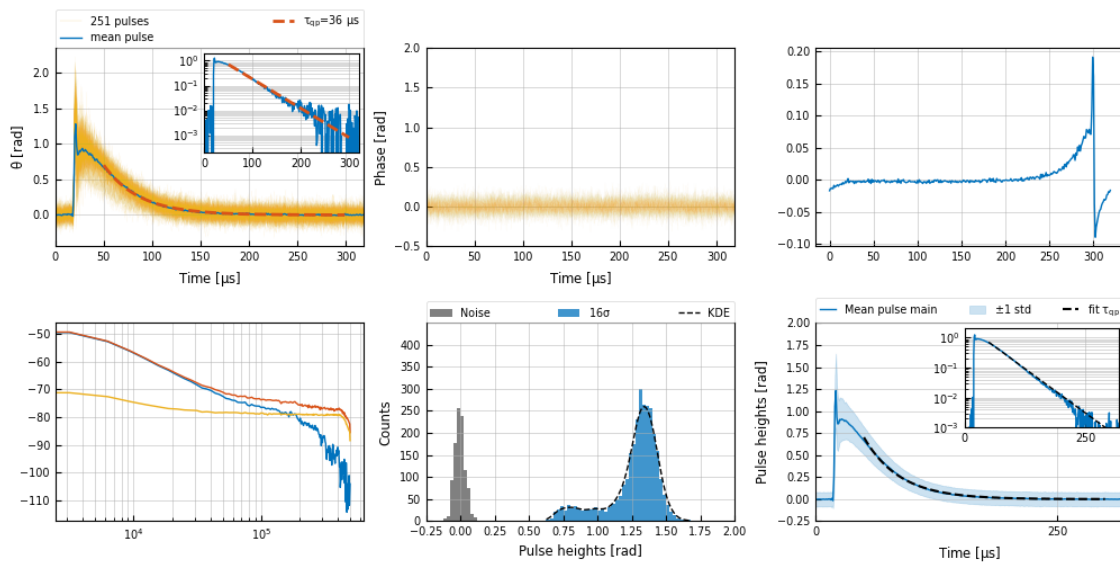
```
fig, axes = plt.subplot_mosaic('abc;def', constrained_layout=True,
figsize=(10, 5))
```

Analysed 300 out of 300 files

Secondaries/Outliers/Total = 356/159/3512

R = 5.93, Rsn = 14.09, 3040 pulses

tqp = 36 +- 0.2 us



```
[54]: kid = 21
pread = 103
file_type = 'vis'
pw = 300
pw_offset = 20
filter_type = 'exp'
lifetime = 25
fit_tqp = [30, 100]

dir_on = r"D:\Data\LT218Chip1_BF_20221103_MIR3_8\12KIDs mono on 3800_
↳long\TD_Power"
chunksize = 25
```

```

noise_mph = 5
pulse_mph = 16
doubles_fraction = 10
nr_stds = [16]
nr_chunks = [None]
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,
    ↪ pulse_mph, pw, pw_offset, lifetime, nr_stds,
    ↪ nr_chunks, chunksize, filter_type, fit_tqp,
    ↪ doubles_fraction=doubles_fraction)

kid_id = np.argwhere(kid_table[:, 1] == kid)[0][0]
name = 'KID%d' % kid_id
pulse_models[name] = {}
pulse_models[name]['pulse template'] = np.mean(pulses, axis=0)

```

Secondaries/Outliers/Total = 154/9/247

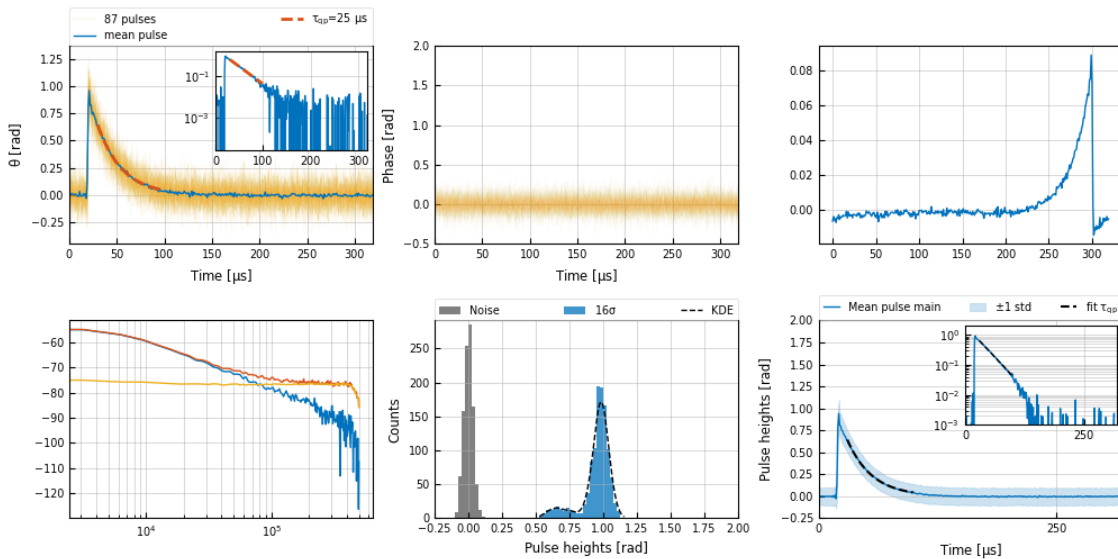
Optimal filter constructed with 87 pulses and 1000 noise segments

Analysed 300 out of 300 files

Secondaries/Outliers/Total = 1842/100/3009

R = 7.26, Rsn = 11.48, 1135 pulses

tqp = 25 +- 0.3 us



```

[55]: kid = 24
      pread = 113
      file_type = 'vis'
      pw = 300
      pw_offset = 20

```

```

filter_type = 'exp'
lifetime = 60
fit_tqp = [50, 300]

dir_on = r"D:\Data\LT218Chip1_BF_20221103_MIR3_8\12KIDs mono on 3800_
↳long\TD_Power"
chunksize = 25
noise_mph = 5
pulse_mph = 16
doubles_fraction = 2
nr_stds = [16]
nr_chunks = [None]
pulses = peaks_vs_thresholds(dir_on, kid, pread, file_type, noise_mph,
↳pulse_mph, pw, pw_offset, lifetime, nr_stds,
nr_chunks, chunksize, filter_type, fit_tqp,
↳doubles_fraction=doubles_fraction)

kid_id = np.argwhere(kid_table[:, 1] == kid)[0][0]
name = 'KID%d' % kid_id
pulse_models[name] = {}
pulse_models[name]['pulse template'] = np.mean(pulses, axis=0)

```

Secondaries/Outliers/Total = 1/28/544

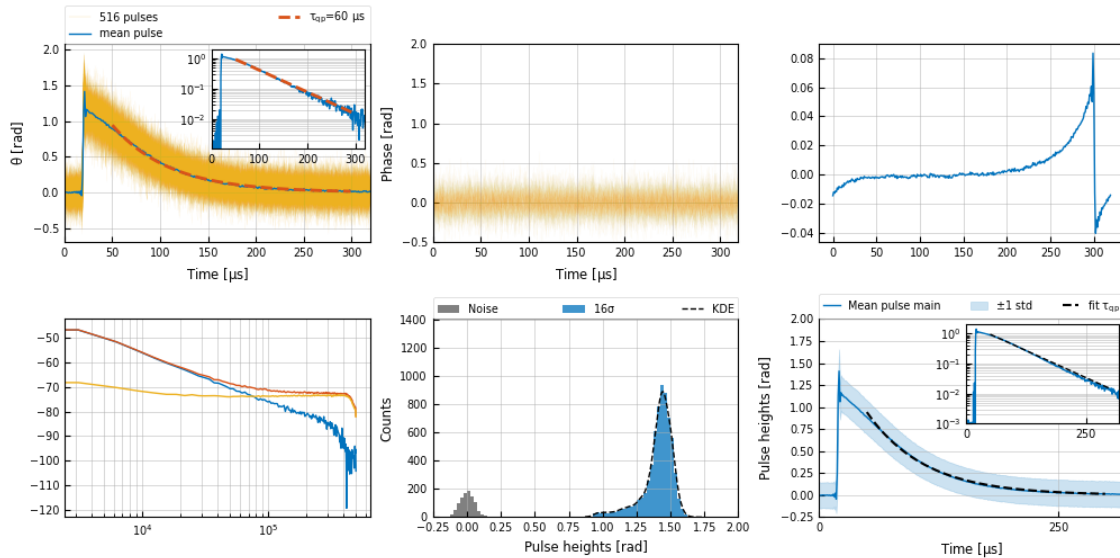
Optimal filter constructed with 516 pulses and 1000 noise segments

Analysed 300 out of 300 files

Secondaries/Outliers/Total = 8/313/7049

R = 9.27, Rsn = 11.75, 6736 pulses

tqp = 60 +- 0.3 us



```
[60]: import pickle

# BEGIN: Save pulse_models as a pickle file
with open('LT218_MUX_darkcounts.pkl', 'wb') as f:
    pickle.dump(pulse_models, f)
# END: Save pulse_models as a pickle file
```